



slow: $1 + t$ $\rightarrow$ location
 $\rightarrow$ take t step

fast: $1 + 2t$ $\rightarrow$ take t step

the first time met in the cycle

$$m-1 + nk_1 + \underset{\Delta}{i} = 1 + t$$

$$m-1 + nk_2 + i = 1 + 2t$$

$$\Rightarrow t = n(k_2 - k_1) \quad \text{Since } t \geq m-1$$

$$n(k_2 - k_1) \geq m-1$$

$$\Rightarrow k_2 - k_1 \geq \frac{m-1}{n}$$

$$\Rightarrow m-1 + nk_1 + i = 1 + n(k_2 - k_1)$$

$$\Rightarrow i = 2 + n(k_2 - k_1) - nk_1 - m$$

$$\rightarrow \text{let } k_2 - k_1 = \lceil \frac{m-1}{n} \rceil = c$$

$$\text{let } k_1 = 0$$

$$\Rightarrow i = 2 + nc - m \rightarrow k_2 = c$$

$$\Rightarrow i = 2 + n \lceil \frac{m-1}{n} \rceil - m$$

eg: $m=10$
 $n=7$ $\rightarrow i = 2 + 7 \times 2 - 10 = 6$
 $\rightarrow$ go additional $m-1$ steps

$$2 + n \lceil \frac{m-1}{n} \rceil - m + \boxed{m-1}$$

important: $\star$
 $= \frac{1}{n} + \frac{n \lceil \frac{m-1}{n} \rceil}{n}$
 $\rightarrow$ go back to the start point of the cycle, an integer about n